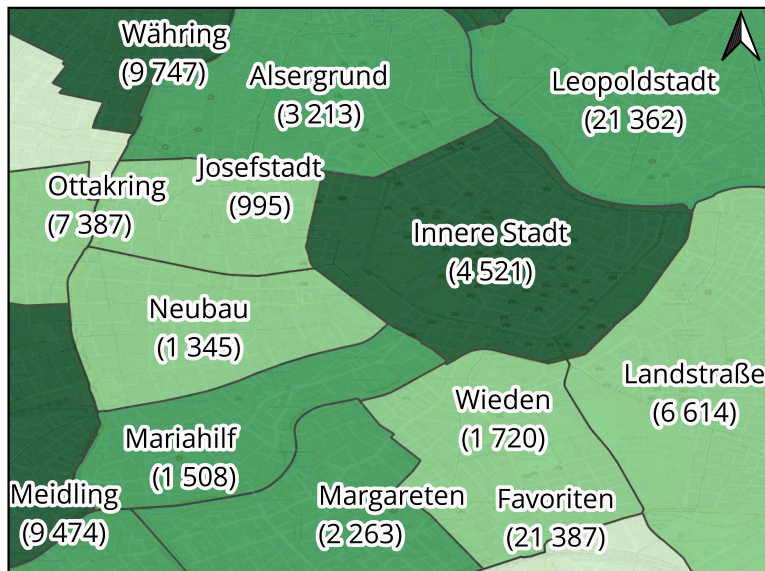
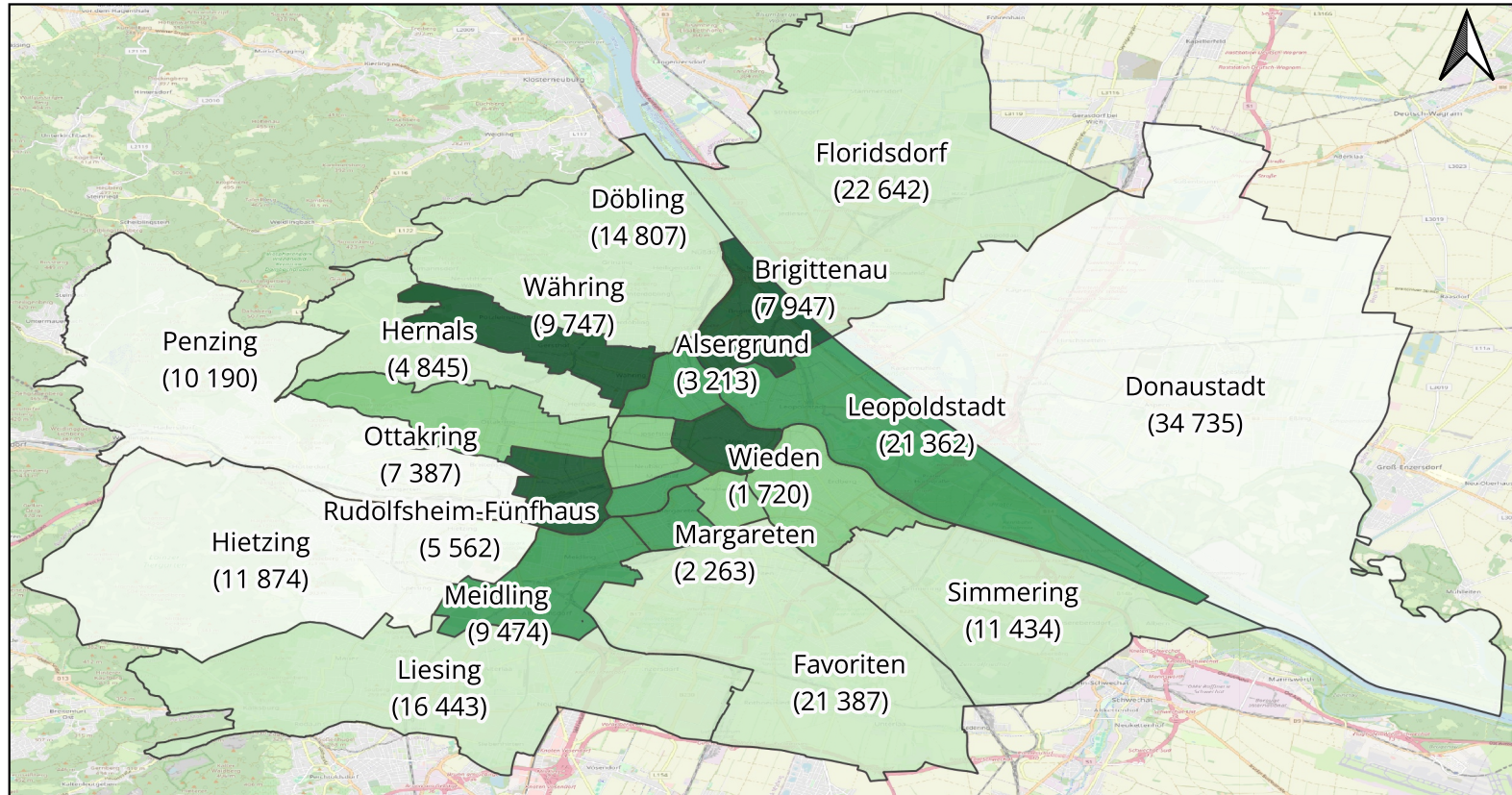


VIENNA | urban tree density - spatial distribution of registered trees across city districts



0 2,5 5 km

1:100 000

TREE DENSITY

- Very low density
- Low density
- Moderate density
- High density
- Very high density

The number in parentheses indicates the absolute count of registered trees in the given district. The high-greenery density index is calculated as the ratio of tree count to the surface area of each administrative unit.

The project was developed in a GIS environment using QGIS and a PostgreSQL / PostGIS spatial database, utilizing two official spatial datasets from Stadt Wien Open Data.

The spatial analysis based on the open data register across Vienna's 23 districts reveals a clear spatial paradox, where high relative greenery density does not directly reflect total tree abundance. Central and inner-city districts such as Brigittenau, Währing, Innere Stadt, and Meidling exhibit the highest relative density values (dark green on the choropleth map) primarily due to their small total surface area, where even a moderate tree count produces a high ratio per square kilometer, though this does not eliminate localized urban microclimate challenges. Conversely, high-volume peripheral districts like Donaustadt and Floridsdorf show low relative density despite holding the city's highest absolute tree counts - 34,735 and 22,642 trees respectively, totaling over 57,000 - because their vast administrative boundaries, including farmland, industrial zones, transit hubs, and water bodies, artificially dilute the density metric. Meanwhile, true deficit zones such as Favoriten and Simmering combine low relative density with an urgent need for targeted greening interventions, as their historical industrial character and dense infrastructure create a genuine shortage of public trees and accessible green spaces.

PostgreSQL



0 500 1 000 m

1:30 000